

Loose Dilations for Quasi-Compact Power-Bounded Operators on L^p

Candidate partial answer to arXiv:1106.1513

11 August 2026

Status: candidate partial result, likely valid. The source problem is answered for every quasi-compact operator. The general p -completely polynomially bounded case remains open here.

1 Source problem and result

Fix $1 < p < \infty$ and $p \neq 2$. Arhancet and Le Merdy [1] ask whether every p -completely polynomially bounded operator $T : L^p(\Omega) \rightarrow L^p(\Omega)$ admits a *loose dilation*: do there exist an L^p -space Y , an isomorphism $U : Y \rightarrow Y$ with

$$\sup_{n \in \mathbb{Z}} \|U^n\| < \infty,$$

and bounded maps $J : L^p(\Omega) \rightarrow Y$, $Q : Y \rightarrow L^p(\Omega)$ such that

$$T^n = QU^nJ \quad (n \geq 0)?$$

Such an operator is automatically power bounded. We prove the desired conclusion under quasi-compactness, with U an onto isometry. Recall that T is quasi-compact when its essential spectral radius is less than one.

Theorem 1. *Let $1 < p < \infty$ and let $T : L^p(\Omega) \rightarrow L^p(\Omega)$ be quasi-compact and power bounded. Then T admits a loose dilation for which the dilating operator is an onto isometry. Consequently, the open problem in [1] has an affirmative answer for every quasi-compact p -completely polynomially bounded operator.*

The proof separates the exponentially stable part from the finite-dimensional peripheral part. We first record the stable construction.

2 An explicit shift dilation of a summable stable part

Let $X = L^p(\Omega)$, let $P \in B(X)$ be a projection commuting with T , and put $X_s = PX$. Write $p' = p/(p-1)$.

Lemma 2. *If*

$$\sum_{k=0}^{\infty} \|T^k P\|^{p'} < \infty, \tag{1}$$

then the restriction of T to X_s has an isometric loose dilation inside $\ell^p(\mathbb{Z}; X)$.

Proof. The space $Y_s = \ell^p(\mathbb{Z}; X)$ is itself an L^p -space. Let V be the bilateral right shift,

$$(Vy)_k = y_{k-1} \quad (k \in \mathbb{Z}).$$

It is an onto isometry. Define

$$J_s x = (\dots, 0, Px, 0, \dots),$$

with Px in coordinate zero, and define

$$Q_s(y) = \sum_{k=0}^{\infty} T^k P y_k. \quad (2)$$

The series converges absolutely in norm after scalar Hölder inequality:

$$\sum_{k \geq 0} \|T^k P y_k\| \leq \left(\sum_{k \geq 0} \|T^k P\|^{p'} \right)^{1/p'} \left(\sum_{k \geq 0} \|y_k\|^p \right)^{1/p}.$$

Thus Q_s is bounded. Since $V^n J_s x$ has Px in coordinate n ,

$$Q_s V^n J_s x = T^n P x \quad (n \geq 0).$$

□

3 Peripheral spectral part

Lemma 3. *Suppose T is quasi-compact and power bounded. There is a bounded spectral projection P_u commuting with T such that $X_u = P_u X$ is finite-dimensional, $T|_{X_u}$ is similar to a diagonal onto isometry on some ℓ_r^p , and the complementary restriction $T|_{(I-P_u)X}$ has spectral radius strictly less than one.*

Proof. The quasi-compact spectral decomposition says that the spectrum outside any circle strictly larger than the essential spectral radius consists of finitely many isolated eigenvalues of finite algebraic multiplicity. Hence the peripheral spectrum $\sigma(T) \cap \mathbb{T}$ is finite and its Riesz spectral subspace X_u is finite-dimensional. Power boundedness rules out nontrivial Jordan blocks at eigenvalues of modulus one. Choosing an eigenbasis identifies $T|_{X_u}$ with

$$D = \text{diag}(\lambda_1, \dots, \lambda_r) \quad \text{on } \ell_r^p, \quad |\lambda_j| = 1,$$

which is an onto isometry. The complementary spectrum is a compact subset of the open unit disc, so the complementary spectral radius is less than one. □

4 Proof of the theorem

Proof of Theorem 1. Take P_u from Lemma 3, put $P_s = I - P_u$, and choose an isomorphism $S : X_u \rightarrow \ell_r^p$ satisfying

$$ST|_{X_u} = DS.$$

Since the spectral radius of $T|_{P_s X}$ is less than one, there are $C \geq 1$ and $0 < \rho < 1$ such that

$$\|T^k P_s\| \leq C \rho^k \quad (k \geq 0).$$

In particular, condition (1) holds. Apply Lemma 2 to obtain Y_s , V , J_s , and Q_s .

Let

$$Y = \ell_r^p \oplus_p Y_s, \quad U = D \oplus V.$$

This is an L^p -space and U is an onto isometry. Define

$$Jx = (SP_u x, J_s x), \quad Q(z, y) = S^{-1}z + Q_s y.$$

Both maps are bounded. For every $n \geq 0$,

$$\begin{aligned} QU^n Jx &= S^{-1}D^n SP_u x + Q_s V^n J_s x \\ &= T^n P_u x + T^n P_s x = T^n x. \end{aligned}$$

Thus (Y, U, J, Q) is the required loose dilation. $\square$

In particular, every power-bounded operator on a finite-dimensional L^p -space admits such a dilation, whether or not it is Ritt.

5 Boundary of the argument

The stable lemma applies more generally whenever the powers on a complemented invariant part have $\ell^{p'}$ -summable operator norms, and the peripheral part can be dilated separately. Quasi-compactness supplies exactly this spectral gap.

For an arbitrary power-bounded T , every Abel regularization rT with $0 < r < 1$ satisfies the stable summability condition. However, the norm of the map in (2) is controlled by

$$\left(\sum_{k \geq 0} r^{kp'} \|T^k\|^{p'} \right)^{1/p'},$$

which generally diverges as $r \uparrow 1$. Thus the construction does not give uniformly bounded dilations from which one could take an ultraproduct limit. Removing this blow-up would require genuinely using p -complete polynomial boundedness, not power boundedness alone.

The natural Hilbert-space proof would extend and factor the completely bounded polynomial calculus. For $p \neq 2$, a general Arveson–Wittstock–Hahn–Banach extension theorem for p -operator spaces is false [2]. No special extension result sufficient for the analytic shift algebra was located in the bounded literature search. This is why the present theorem is a scoped answer rather than a claimed general solution.

References

- [1] C. Arhancet and C. Le Merdy, *Dilation of Ritt operators on L^p -spaces*, Israel J. Math. **201** (2014), 373–414; arXiv:1106.1513.
- [2] J.-J. Lee, *Hahn–Banach type extension theorems on p -operator spaces*, Houston J. Math. **41** (2015), 799–814; arXiv:1303.3513.